

Configuring Strata Cloud Manager for an AIRS Firewall Deployment

Student Guide

This guide provides step-by-step instructions for configuring the Strata Cloud Manager (SCM) during an AI Runtime Security (AIRS) firewall deployment in a lab environment.

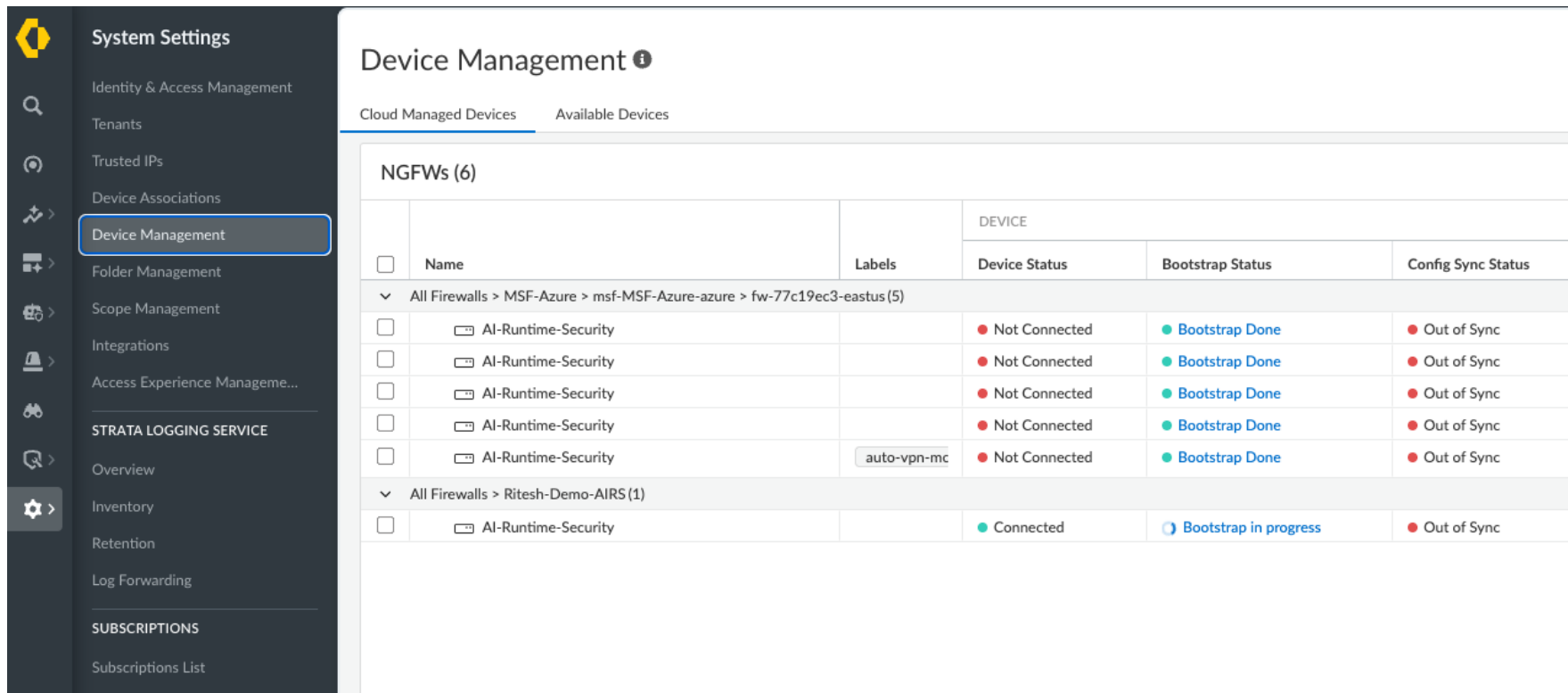
1. Firewall Bootstrapping and Registration

Upon completing the **Phase 2** deployment using the Terraform script, the firewall will initialize its boot sequence.

1. **Automatic Registration:** The firewall will automatically register with the cloud licensing server using the parameters passed during bootup.
2. **Verify Serial Number:** Check your deployment profile to confirm that the firewall's serial number is now visible.

▼	Ritesh_AIRS_FW1	VM-AI	Flexible vCPUs (PAN-OS 11.2.5 and above)	37.34 / 46.67	1 / 1	4 / 4	N/A	 View Devices View Tenant ?
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3. **Monitor Bootstrap Status:** Navigate to **SCM Tenant > System Settings > Device Management**. You will notice the firewall status is currently displayed as *"Bootstrap in progress"*.



	Name	Labels	DEVICE		
			Device Status	Bootstrap Status	Config Sync Status
All Firewalls > MSF-Azure > msf-MSF-Azure-azure > fw-77c19ec3-eastus (5)					
☐	AI-Runtime-Security		Not Connected	Bootstrap Done	Out of Sync
☐	AI-Runtime-Security		Not Connected	Bootstrap Done	Out of Sync
☐	AI-Runtime-Security		Not Connected	Bootstrap Done	Out of Sync
☐	AI-Runtime-Security		Not Connected	Bootstrap Done	Out of Sync
☐	AI-Runtime-Security	auto-vpn-mc	Not Connected	Bootstrap Done	Out of Sync
All Firewalls > Ritesh-Demo-AIRS (1)					
☐	AI-Runtime-Security		Connected	Bootstrap in progress	Out of Sync

Note: The bootstrap process takes approximately 20 minutes to complete. During this time, the firewall installs necessary device content packages and performs a reboot.

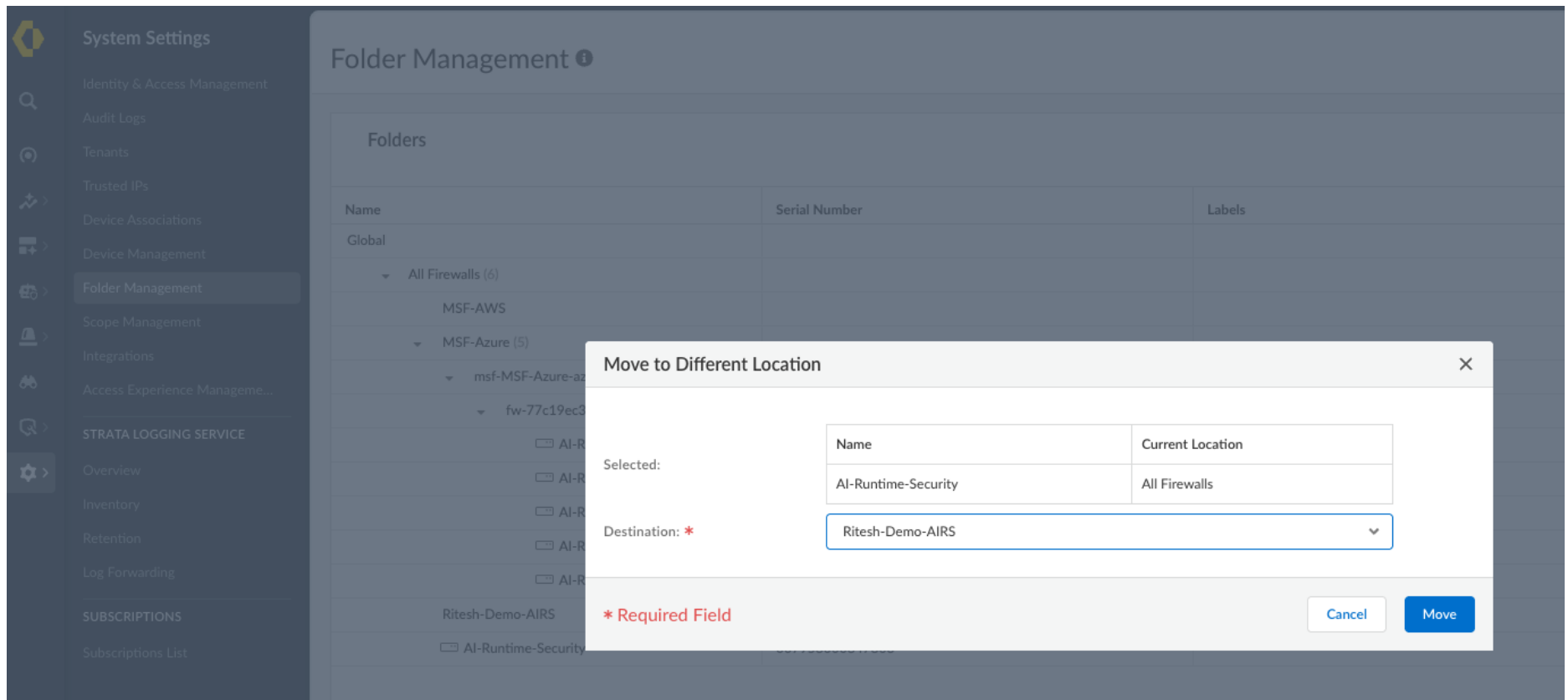
2. Folder Management and Configuration Scope

While the firewall is bootstrapping, you can proceed with the SCM configuration.

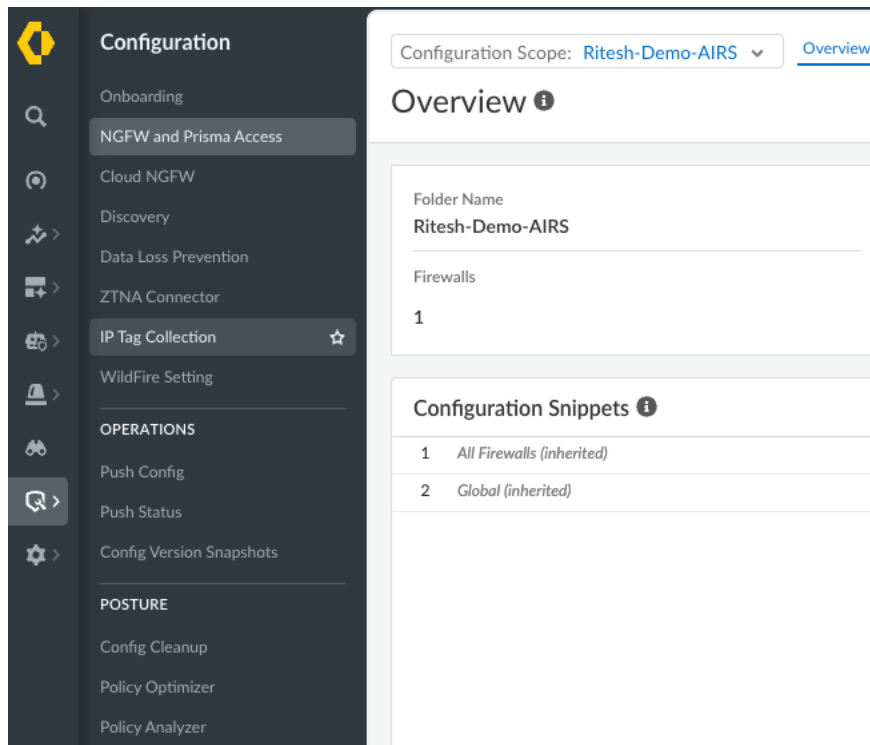
1. **Folder Assignment:** Under **Folder Management**, create a new folder and move your newly deployed firewall into it.

The screenshot displays the 'System Settings' sidebar on the left, with 'Folder Management' selected. The main panel is titled 'Folder Management' and contains a table of folders. The table has four columns: Name, Serial Number, Labels, and Actions. The 'Name' column shows a hierarchical structure starting with 'Global', followed by 'All Firewalls (6)', 'MSF-AWS', 'MSF-Azure (5)', 'msf-MSF-Azure-azure (5)', and 'fw-77c19ec3-eastus (5)'. Under 'fw-77c19ec3-eastus (5)', there are five entries for 'AI-Runtime-Security' with serial numbers 007957000843131, 007957000843132, 007957000845854, 007957000845855, and 007957000846537. The last entry has a label 'auto-vpn-mcn-azure-eastus'. Below these is 'Ritesh-Demo-AIRS' with an 'AI-Runtime-Security' entry with serial number 007958000848622. The 'Actions' column contains vertical ellipses for each row. At the bottom right, there are 'Edit' and 'Move' buttons.

Name	Serial Number	Labels	Actions
Global			
▼ All Firewalls (6)			⋮
MSF-AWS			⋮
▼ MSF-Azure (5)			⋮
▼ msf-MSF-Azure-azure (5)			⋮
▼ fw-77c19ec3-eastus (5)			⋮
AI-Runtime-Security	007957000843131		⋮
AI-Runtime-Security	007957000843132		⋮
AI-Runtime-Security	007957000845854		⋮
AI-Runtime-Security	007957000845855		⋮
AI-Runtime-Security	007957000846537	auto-vpn-mcn-azure-eastus	⋮
Ritesh-Demo-AIRS			⋮
AI-Runtime-Security	007958000848622		⋮



2. **Set Configuration Scope:** Navigate to **Configuration > NGFW and Prisma Access**. Change the configuration scope at the top of the screen to the folder you just created.



3. Device Configuration

We will adopt a right-to-left configuration flow, beginning with the **Device** settings.

3.1. Zones and Interfaces

1. **Create Zones:** Create the following zones:
 - o Untrust

- o Trust

(Optional: You can create a 'Zone Protection' profile and attach it to these zones. For this lab, we will proceed with the default settings).

Configuration Scope: [Ritesh-Demo-AIRS](#) Overview Security Services Network Policies Identity Services Objects [Device](#) Operations

Zones ⓘ

Zones (9)

☐	Name			Location	Type	Interfaces	Zone Protec...	Packet Buffe...	USER-ID	
									Enabled	Includ
☐	zone-internal			All Firewalls (Auto-VPN-De...)	layer3			☒	☐	
☐	zone-to-hub			All Firewalls (Auto-VPN-De...)	layer3			☒	☐	
☐	zone-to-branch			All Firewalls (Auto-VPN-De...)	layer3			☒	☐	
☐	zone-to-pa-hub			All Firewalls (Auto-VPN-De...)	layer3			☒	☐	
☐	local			All Firewalls	layer3	\$eth-local	best-practice	☒	☒	
☐	internet			All Firewalls	layer3	\$eth-internet	best-practice	☒	☐	
☐	proxy			All Firewalls	layer3			☒	☐	
☐	Untrust			Ritesh-Demo-AIRS	layer3	\$eth-untrust		☒	☐	
☐	Trust			Ritesh-Demo-AIRS	layer3	\$eth-trust		☒	☐	

2. **Create Interfaces:** Configure the **Untrust** and **Trust** interfaces.
 - You can assign static IP addresses manually or use an **Address Object** (see the *Objects Configuration* section below).

Configuration Scope: [Ritesh-Demo-AIRS](#) Overview Security Services Network Policies Identity Services Objects **Device** Operations

Interfaces ?

[Push Config](#)

Ethernet VLAN Loopback Tunnel Cellular

Ethernets (4) [Delete](#) [Override](#) [Add Interface](#)

☐	Interface	Location	Interface Type	IP Address	Security Zone	Logical Router	Tag	Management Profile	Comment
▼ Interfaces (4)									
☐	\$eth-internet	All Firewalls	Layer 3	None	internet	default	Untagged	None	Internet Interface
☐	\$eth-local	All Firewalls	Layer 3	None	local	default	Untagged	None	Internal Interface
☐	\$eth-untrust	Ritesh-Demo-AIRS	Layer 3	untust-int	Untrust	airs-logical-router	Untagged	None	eth1/1
☐	\$eth-trust	Ritesh-Demo-AIRS	Layer 3	trust-int	Trust	airs-logical-router	Untagged	enable_ping	eth1/2

3.2. Logical Router and Routing

1. **Create Logical Router:** Navigate to the Logical Router section, create a new router, and assign the previously created Untrust and Trust interfaces to it.
2. **Configure Static Route:** Within the Logical Router, create a static default route directing all outbound traffic through the Untrust interface.
 - **Destination:** 0.0.0.0/0
 - **Next Hop IP Address:** 172.16.1.1

[Routers](#) [[Global](#) / [All Firewalls](#) / [Ritesh-Demo-AIRS](#)] > [Routers](#)

airs-logical-router

General

Name * ECMP ☐ Enable

Interfaces (2)

☐ Interface Name☐ \$eth-untrust☐ \$eth-trust

ADVANCED SETTINGS ▾

Static Route

IPv4 Static Routes 1

IPv6 Static Routes 0

Static Route

IPv4s (1)										
Search Delete										
☐	Name	Destination	Interface	Next Hop	Admin Dist	Metric	BFD	Path Monit...	Failure Con...	Preemptive...
☐	default-route	0.0.0.0/0	\$eth-untrust	ip_address		10	None			

4. Objects Configuration

Navigate to the **Objects** section to define the necessary address objects and groups.

1. **Address Objects:** Create the following address objects (given in the screenshot below) (The one's ending with 'int' can be referenced directly in the interface static IP configuration (above)):

- o trust-int - 172.16.2.10/24
- o untrust-int - 172.16.1.10/24
- o Trust_Subnet - 172.16.2.0/24
- o VertexAI-Gen - generativelanguage.googleapis.com
- o VertexAI-USCentral - us-central1-aiplatform.googleapis.com

☐	Name	Security Post...	Tags	Location	Days Unused	Description	Type	
☐	Palo Alto Networks...			Global (Global-Default)		Palo Alto Network...	fqdn	sinkhole.paloaltonetworks.com
☐	Trust_Subnet			Ritesh-Demo-AIRS			ip-netmask	172.16.2.0/24
☐	untust-int			Ritesh-Demo-AIRS			ip-netmask	172.16.1.10/24
☐	trust-int			Ritesh-Demo-AIRS			ip-netmask	172.16.2.10/24
☐	VertexAI-Gen			Ritesh-Demo-AIRS			fqdn	generativelanguage.googleapis.com
☐	VertexAI-USCentral			Ritesh-Demo-AIRS			fqdn	us-central1-aiplatform.googleapis.com

2. **Address Group:** Create an Address Group containing the Fully Qualified Domain Names (FQDNs) for **VertexAI-Decrypt**.
Address Groups

Address Groups (1)							
						Search	
☐	Name	Security Postu...	Tags	Location	Members Count		Da
☐	VertexAI-Decrypt			Ritesh-Demo-A...	2	VertexAI-Gen VertexAI-USCentral	

5. Network Policies (Source NAT)

Moving to the **Network Policies** section, configure the Source NAT to ensure outbound traffic is translated correctly.

1. Create a Source NAT policy rule for traffic exiting the `Untrust` interface.

NAT ⓘ Push Config ▾

NAT Policy Rules (1)

Filter
Grid
Delete
Troubleshooting
Enable
Disable
Clone
Move ▾
Add Rule
Settings

	#	Name	Tag	ORIGINAL PACKET						TRANSLATED PACKET
				Zone	Address	Destination Zone	Destination Interf...	Destinati...	Service	Source Translation
▼ Ritesh-Demo-AIRS - Pre Rules (1)										
☐	1	Trust-to-UnTrust-SNAT		Trust	Trust_Subnet	Untrust	\$eth-untrust	any	any	dynamic-ip-and-port: interface-address: interface: \$eth-untrust, ip: untust-int

Trust-to-UnTrust-SNAT

Created: 2026-05-01 | Last Modified: 2026-05-01 12:25pm

Original Packet

Source

ZONES * Custom ▾

Zones

Trust × +

ADDRESSES * Custom ▾

Addresses

Trust_Subnet × +

[Add Address Groups](#)[Add External Dynamic Lists](#)

Destination

ZONE *

Untrust ▾

INTERFACE

\$eth-untrust (ethernet1/1) ▾

ADDRESSES * Any Address

[Add Addresses](#)[Add Address Groups](#)[Add External Dynamic Lists](#)[Add SaaS Application Endpoints](#)

Service *

any

Translated Packet

Determine the type of translation to perform on the source address, destination address, or for both source and destination addresses.

Translation

☐ None ☒ Source Address Only ☐ Destination Address Only ☐ Both

Source Address Translation

Translation Type

☒ Dynamic IP and Port ☐ Dynamic IP ☐ Static IP

☐ Persistent NAT

Choice

☐ Translated Address ☒ Interface Address

Interface Address

Interface *

\$eth-untrust (ethernet1/1)

Choice

☒ IP ☐ Floating IP

IP

untust-int

6. Security Services and Profiles

Next, move further left to the **Security Services** section.

6.1. AI Security Profile

Ritesh Tandon (Sr. Technical Marketing Engineer)

To test the AI application, we need a profile that governs how the firewall handles LLM prompts and responses.

1. **Create Custom Profile:** Create a custom **AI Security Profile** for this lab. This profile dictates whether the firewall will allow, alert, or block specific AI requests and LLM responses.

Configuration Scope: [Ritesh-Demo-AIRS](#) ▼ Overview [Security Services](#) ▼ Network Policies ▼ Identity Services ▼ Objects ▼ Device ▼ Operations ▼ 📌

AI Security ⓘ

AI Security Profiles (2)					🔍 Search	🔼
☐	Name		Security Posture	Location	Profile Groups	
☐	best-practice	🔒	🔄	Global (predefined)		
☐	AIRS-Custom-Profile		🔄	Ritesh-Demo-AIRS	AI-Security-Traffic	

AIRS-Custom-Profile

Configuration Profile Usage

Security Rules Using This Profile 2

Profile Groups Containing This Profile 1

Model Groups (1)

The default group applies access control and security settings for all models. To edit settings for specific models, add a new model group.

Delete

Add Model Group

	Name	Access Control	Target Models	REQUEST PROTECTION			RESPONSE PROTECTION		
				AI Model	AI Application	AI Data	AI Model	AI Application	AI Data
☐	default	☒ Allow	All	Prompt Injection Block	URL Security Default: Block Allow Category: 0 Alert Category: 3 Block Category: 13	Data Rule PII	Toxic Content High: Block Moderate: Block	URL Security Default: Block Allow Category: 0 Alert Category: 3 Block Category: 13	Data Rule PII
				Toxic Content High: Block Moderate: Block				Database Security Create: Alert Read: Allow Update: Block Delete: Block	

Advanced Settings

Latency

Max Inline Latency SecInline Timeout Action ☒ Allow ☐ Alert (Report Threats Asynchronously) ☐ BlockCustom Model Support ☒ Enable

All traffic matching this security profile will be forwarded to the Palo Alto Networks AI security cloud service for threat inspection, rather than just traffic to known model endpoints.

2. **Security Profile Group:** Create a **Security Profile Group**. Tie your new custom AI Security Profile into this group alongside the default *Best Practice* security profiles.

[Profile Group](#) [Global / All Firewalls / Ritesh-Demo-AIRS] > Profile Group - Ritesh-Demo-AIRS

AI-Security-Traffic

Configuration Usage

Profile Group	
Name *	AI-Security-Traffic
Anti-Spyware Profile	best-practice X v
Vulnerability Protection Profile	best-practice X v
URL Access Management Profile	best-practice X v
File Blocking Profile	best-practice X v
HTTP Header Insertion Profile	None v
WildFire and Antivirus Profile	best-practice X v
DNS Security Profile	None v
AI Security Profile	AIRS-Custom-Profile X v
Data Loss Prevention Profile	None v

7. Security Rules

We need to establish rules to govern traffic flows from the AI Application server. Create the following two security rules:

1. **AI Traffic Rule:** Allow traffic from the *AI Application Server* to the *Vertex AI* endpoints.
2. **Standard Outbound Rule:** Allow standard internet traffic (HTTPS, DNS, NTP) from the *AI Application Server* to the internet.

Ritesh-Demo-AIRS - Pre Rules (2)													
☐	2	 Trust-to-Internet	any	  	 Trust	 Trust_Subnet	any	any	any	any	any	 Untrust	any
☐	3	 Trust-to-VertexAI	any	  	 Trust	 Trust_Subnet	any	any	any	any	any	 Untrust	any

[Security Policy](#) [Global / All Firewalls / Ritesh-Demo-AIRS] > Security Policy

Trust-to-Internet

Created: 2026-05-01 | Last Modified: 2026-05-18 10:41am

Match Criteria

SOURCE

Zones (1) ☐ Any ☒ Select

Trust X

Addresses (1) ☐ Any ☒ Select

Trust_Subnet X

☐ NegateUsers ☒ Any ☐ Select ☐ Pre Logon ☐ Known User ☐ UnknownDevices ☒ Any ☐ Select ☐ No-hip ☐ Quarantined Devices

APPLICATION / SERVICE

Application (4) ☐ Any ☒ Select

ssl X web-browsing X ntp-base X dns-base X

Service ☒ Application Default ☐ Any ☐ Select

DESTINATION

Zones (1) ☐ Any ☒ Select ☐ Multicast

Untrust X

Addresses ☒ Any ☐ SelectDevices ☒ Any ☐ Select ☐ Quarantined Devices

URL CATEGORY / TENANT RESTRICTION

URL Category ☒ Any ☐ SelectTenant Restriction ☒ Any ☐ Select

Actions

Action

Allow X

Profile Group

Log Settings

Others

Profile Group

AI-Security-Traffic >

Trust-to-VertexAI

Created: 2026-05-01 | Last Modified: 2026-05-18 10:41am

SOURCE		DESTINATION	
Zones (1) ☐ Any ☒ Select Trust ✕		Zones (1) ☐ Any ☒ Select ☐ Multicast Untrust ✕	
Addresses (1) ☐ Any ☒ Select Trust_Subnet ✕ ☐ Negate		Addresses ☒ Any ☐ Select Devices ☒ Any ☐ Select ☐ Quarantined Devices	
Users ☒ Any ☐ Select ☐ Pre Logon ☐ Known User ☐ Unknown Devices ☒ Any ☐ Select ☐ No-hip ☐ Quarantined Devices			
APPLICATION / SERVICE Application (2) ☐ Any ☒ Select google-base ✕ google-vertex-ai-base ✕ Service ☒ Application Default ☐ Any ☐ Select		URL CATEGORY / TENANT RESTRICTION URL Category ☒ Any ☐ Select Tenant Restriction ☒ Any ☐ Select	
Actions Action Allow ✕ ▼		Profile Group Log Settings Others Profile Group AI-Security-Traffic ✕ ▼	

8. Decryption Policies

For the firewall to gain visibility into the context (requests and responses) of the LLM traffic, it must decrypt the connection between the AI Application Server and Vertex AI.

1. Create Decryption Rules:

- **Decrypt-Trust-to-VertexAI:** Create a policy to decrypt traffic originating from the *AI Server* destined for *Vertex AI*.
- **No-Decryption:** Create a secondary policy to bypass decryption for all other general internet-bound traffic.

Decryption ⓘ

Push

Decryption Policies (2)

Decrypt SSL traffic, including SSL encapsulated protocols, for visibility, control, and granular security.

Search

Delete

Enable

Disable

Clone

Move

Add Rule

☐	#	Name	Description	Secu...	Source Z...	Source A...	Source U...	Destinati...	Destination A...	URL Cate...	Action	Profile	Type
▼ Ritesh-Demo-AIRS - Pre Rules (2)													
☐	1	Decrypt-Trust-to-VertexAI			Trust	Trust_Subnet	any		VertexAI-Decrypt	any	decrypt	best-practice	ssl-forward-proxy
☐	2	No-Decrypt-Others			any	any	any	any	any	any	no-decrypt		ssl-forward-proxy

- ### 2. Map Certificates:
- Within the Decrypt-Trust-to-VertexAI decryption profile, map the necessary certificates utilized for ssl-forward-proxy traffic.

Decryption Settings

Best Practice Check

Certificate Settings

* Certificate When Proxying for Trusted Sites

RSA

Forward-Trust-CA

Validuntil Apr 12 16:44:37 2028 GMT

ECDSA

Forward-Trust-CA-ECDSA

Validuntil Apr 12 16:44:38 2028 GMT

Certificate When Proxying for Untrusted Sites

RSA

Forward-UnTrust-CA

Validuntil Apr 12 16:44:38 2028 GMT

ECDSA

Forward-UnTrust-CA-ECDSA

Validuntil Apr 12 16:44:39 2028 GMT

Inspection Settings

Present response pages even while not decrypting URLs

Enabled

Inspect TLS handshake message

Disabled

End of Lab Configuration Guide